Beam-induced background rejection with hit timing and direction in the barrel tracker of a muon collider, and the resolutions it requires

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Abstract: At a 10 TeV muon collider the decays of the circulating muons fill the tracker with millions of background hits per bunch crossing. We study how far this beam-induced background can be reduced hit by hit, before any pattern recognition, if every hit carries a time and a direction measurement, and what that implies for track reconstruction in the barrel of the inner and outer trackers. Three cuts test whether a hit is compatible with a particle coming from the interaction point: on its time corrected for the time of flight, on the extrapolation of its direction to the beam axis, and on the transverse momentum implied by its direction. With a position resolution of 100 μm, a time resolution of 100 ps and a direction resolution of 1.5°, and cuts set to keep 98% of the signal hits each, the background in the six barrel layers is reduced by a factor of 57 while 97–98% of the hits of signal muons and 98% of the tracks are kept. The algorithm-independent lower limit on combinatorial fake tracks of a companion paper then gives 1.7×10^{-2} fakes per bunch crossing in the whole barrel, a factor 2.3 below the density at which one fake per crossing is expected. Scans of the three resolutions show how this margin depends on each of them and set targets for detector R&D.

Keywords: Pattern recognition, cluster finding, calibration and fitting methods; Performance of High Energy Physics Detectors; Timing detectors; Particle tracking detectors

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1 Introduction

A multi-TeV muon collider would combine the discovery reach of a hadron collider of much higher energy with the clean final states of lepton collisions [1, 2]. The price, for the detector, is the beam-induced background (BIB): the decay products of the circulating muons interact with the machine elements near the interaction region and reach the detector as a diffuse flux of low-energy photons, electrons, neutrons and charged hadrons, out of time and out of direction with respect to the collisions [3, 4, 5]. In the tracker this flux produces millions of hits per bunch crossing, with densities above one hit per mm^2 in the innermost layers, and the reconstruction of charged tracks among them is the central challenge of the tracker design.

Several handles against the BIB are known. The hits arrive spread over a few nanoseconds while those of particles from the collision arrive within tens of picoseconds of the expected time of flight, so a timing window removes much of the background [3, 5]. The BIB particles also cross the sensors at directions unrelated to the interaction point, which double-layer sensors or cluster shapes can exploit. Full reconstruction chains built on these ideas, and the detector concepts proposed for a 10 TeV collider [6, 7], show that tracking is possible, with rates of fake tracks that depend on the algorithm used.

This paper asks a complementary question. If every hit carries a time and a direction, how much of the BIB can be removed hit by hit, before any pattern recognition, while keeping nearly all the hits of particles from the interaction point; and does what remains still allow tracks to be found by *any* algorithm? The first part is a selection on three per-hit observables, each testing the compatibility of the hit with a particle produced at the interaction point. The second uses the algorithm-independent lower limit on combinatorial fake tracks derived in a companion paper [8]: the expected number of combinations of background hits that pass the same goodness-of-fit cut as real tracks, which no algorithm keeping the efficiency for real tracks can reject. Both are evaluated for the six barrel layers of the inner and outer trackers, where the limit can be computed for the whole barrel from the measured hit densities. Finally, scanning the position, time and direction resolutions shows how much margin the selection leaves and which resolution matters most. The direction measurement in particular is treated as a capability to be developed: the scans give the precision it would need.

Section 2 describes the simulated samples and the detector, section 3 the per-hit observables and section 4 the selection and its performance. Section 5 applies the fake-track limit to the barrel, section 6 presents the resolution scans and section 7 discusses the limitations.

2 Simulated samples and detector

Detector. We use the MUSIC detector concept for a 10 TeV muon collider [6] [confirm the geometry version and the reference], with a 5 T solenoidal field. Its tracker consists of a vertex detector (VXD), an inner tracker (IT) and an outer tracker (OT), each with a barrel and two endcaps. This paper concentrates on the barrel of the IT and OT: three IT layers at radii of 165, 355 and 555 mm and three OT layers at 820, 1154 and 1491 mm, with the half-lengths and sensitive areas listed in table 1. The cuts of section 4 are derived for every subsystem, but the results quoted are for these six layers. The sensitive areas are taken from the geometry description of the modules.

Background sample. One bunch crossing of BIB, the sum of the samples of the μ^+ and μ^- beams and of the incoherent pair production at the interaction point, [provenance: generation with FLUKA, detector simulation; who produced it and which version], gives 17.3 million hits in the tracker, of which 7.7 million are in the six barrel layers. Each hit has its position, its local sensor coordinates, its time and the momentum of the particle that produced it, from which its direction is computed.

Signal sample. 100 000 single muons, with transverse momenta from 1.5 GeV/c upwards, polar angles between 10° and 170°, full azimuth and a longitudinal vertex spread of 1.5 mm. Only the hits of the muon itself are used; those of secondary particles (delta rays) are left out. For the efficiency for high-momentum tracks we extrapolate to $p_T \to \infty$ (appendix A).

Resolutions. The simulated hits are exact. We add Gaussian smearing to represent the detector: $\sigma = 100$ μm on both local coordinates, $\sigma_t = 100$ ps on the time and $\sigma_\alpha = 1.5°$ on each of the two angles that define the hit direction relative to the sensor. The same smearing is applied to signal and background. The position and time values are conventional for present-day pixel detectors with timing; a per-hit direction measurement of 1.5° does not exist today in a single sensor and is used here as a working hypothesis, whose consequences section 6 quantifies.

Table 1: The six barrel layers: mean radius, half-length, sensitive area, and the mean BIB hit density before and after the selection of section 4.

layer	R (mm)	$\lvert z\rvert_{\max}$ (mm)	area (m^2)	density (mm^{-2}) before	density (mm^{-2}) after	rejection
IT L0	165	482	1.04	1.58	0.0392	40
IT L1	355	482	2.20	0.552	0.0112	49
IT L2	555	692	5.17	0.244	0.00373	66
OT L0	820	1264	14.0	0.101	0.00131	77
OT L1	1154	1264	19.5	0.0603	0.000837	72
OT L2	1491	1264	28.6	0.0356	0.000541	66

3 Per-hit observables

A particle from the interaction point reaches each layer at a predictable time and with a direction that points back to the beam line. Three observables, computed for each hit from its position, time and direction alone, measure how far the hit is from this hypothesis.

Longitudinal intercept z_0. In the meridian plane containing the beam axis and the hit, the hit direction defines a straight line through the hit. Its intersection with the beam axis is

$$z_0 = z - r\,\frac{p_z}{p_r}, \tag{1}$$

where r and z are the hit coordinates and p_r, p_z the radial and longitudinal components of the direction. In a solenoidal field the trajectory projected on the meridian plane is a straight line, so a particle from the interaction point has z_0 within the luminous region, smeared by the direction resolution.

Transverse momentum from the direction. In the transverse plane, the circle through the origin and the hit, tangent at the hit to the transverse direction, has curvature

$$\frac{1}{R} = \frac{2\sin\psi}{r}, \tag{2}$$

where ψ is the angle between the transverse direction and the radial direction. It is the curvature a particle from the beam line would need to reach the hit in that direction, and gives the transverse momentum $p_T = 0.3\,B\,R$ (GeV/c, T, m). The cut is on $|1/R|$, which is finite and continuous at infinite momentum.

Corrected time t_{corr}. The time of flight of a muon from the interaction point to the hit along that helix, $s/(\beta c)$, is subtracted from the hit time. The path length s is the arc of the circle above divided by $\sin\vartheta$, the polar angle taken from the hit direction, and the speed β follows from the momentum estimated from the curvature and the muon mass. For a muon from the collision t_{corr} is zero within the time resolution; for a background hit it carries the delay of the background particle.

Distributions. Figure 1 shows the three observables in the IT and OT barrel for background and signal hits, each after the cuts on the other two observables. The signal peaks sharply: t_{corr} within the time resolution, z_0 within a few centimetres in the IT and about a decimetre in the OT, where the lever arm of the direction measurement is longer, and the curvature within the range of the muon momenta. The background is broad in all three. Its corrected time peaks about 0.3–0.5 ns late, with a long tail of delayed hits and a smaller population arriving early. [the origin of the late peak is not yet understood; Monte Carlo truth may explain it]

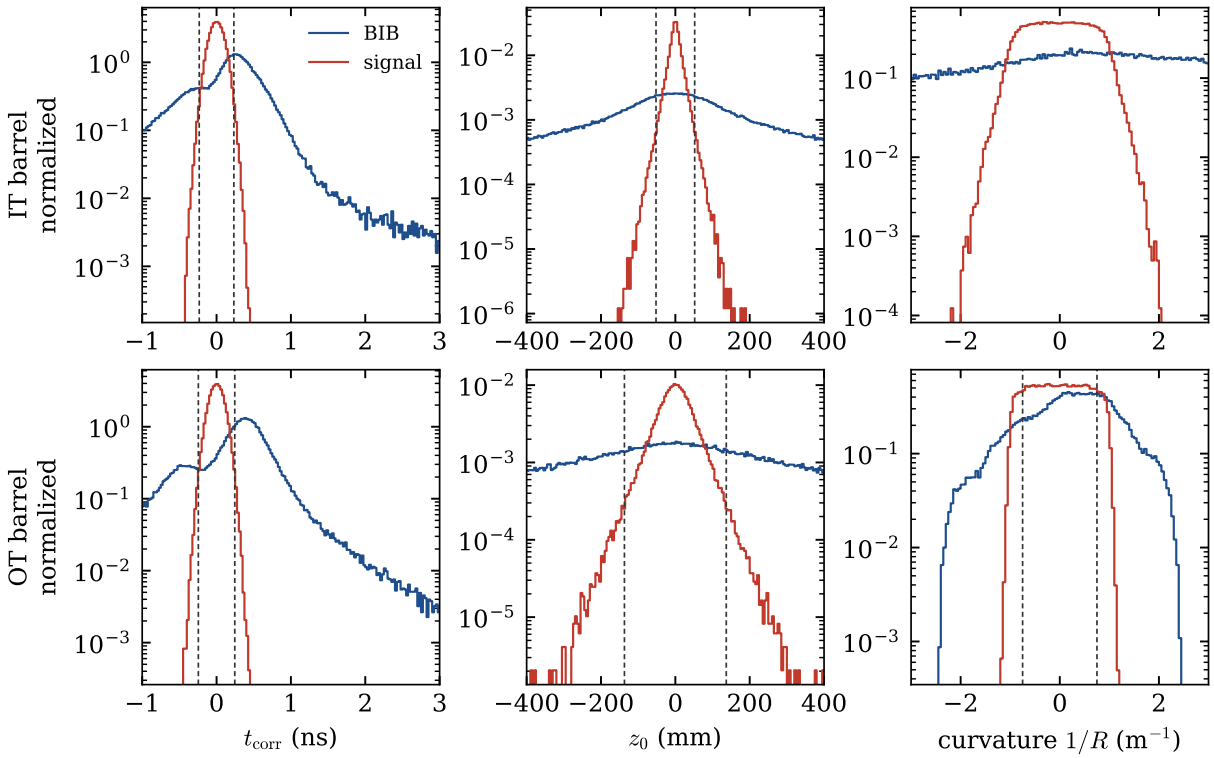


Figure 1: Corrected time, longitudinal intercept and curvature of the hits in the IT barrel (top) and OT barrel (bottom), for one bunch crossing of background (blue) and for signal muons (red), each normalized to unit area. Each distribution is shown after the cuts on the other two observables (N−1); the dashed lines are the cuts of table 2. No curvature cut is applied in the IT.

4 Selection

Construction of the cuts. Each subsystem has a symmetric window on t_{corr} and on z_0, and the OT has in addition a minimum transverse momentum of 2 GeV/c (a maximum $|1/R|$). The time and z_0 windows are set to keep 98% of the signal hits in their N−1 distributions, that is among the hits passing the other cuts. Since each window depends on the other, they are found together by iteration, which converges in two or three steps. The cuts are thus fixed by

the signal alone, not tuned on the background, and they follow automatically when a resolution changes. Table 2 lists them for the barrel.

Table 2: Cuts, background rejection factor and signal hit efficiency in the IT and OT barrel. Rejection factor: hits before over hits after the cuts. Efficiency: fraction of the muon's own hits kept, extrapolated to $p_T \to \infty$. "Alone": each cut applied alone; "N−1": each cut applied on top of the other two.

	IT barrel			OT barrel		
	cut	rejection	efficiency	cut	rejection	efficiency
$\lvert t_{\text{corr}}\rvert$	0.233 ns	10.3 (4.6)	98.4%	0.245 ns	10.2 (5.8)	98.7%
$\lvert z_0\rvert$	52 mm	10.5 (4.7)	98.4%	137 mm	7.8 (3.6)	99.4%
p_T	—	—	—	2 GeV/c	1.8 (1.8)	100%
all		48.6	96.9%		72.0	98.1%

Rejection factors: each cut alone (in parentheses: on top of the other two).

Background rejection. Together the cuts reduce the background in the six barrel layers from 7.7 million to 135 000 hits per bunch crossing, a factor of 57: 49 in the IT barrel and 72 in the OT barrel. The time and z_0 cuts are comparably powerful, a factor of about ten each alone and four to six on top of the others; the momentum cut adds a factor of two in the OT. The densities after the cuts range from 3.9×10^{-2} mm^{-2} in the innermost IT layer to 5.4×10^{-4} mm^{-2} in the outermost OT layer (table 1, figure 2).

Signal efficiency. The cuts keep 96.9% of the hits of high-momentum muons in the IT barrel and 98.1% in the OT barrel. A track is counted as found if at least five of its hits in the IT and OT survive (vertex-detector hits are not counted); the track-finding efficiency for $p_T \to \infty$ is $97.9 \pm 0.2\%$ (figure 3), and $98.1 \pm 0.3\%$ for muons confined to the barrel. It falls below about 2.5 GeV/c, where the momentum cut of the OT and the curvature of the track remove hits.

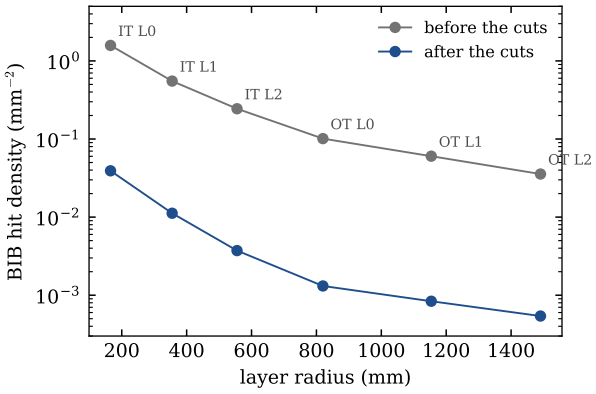


Figure 2: Mean BIB hit density in the six barrel layers, before and after the cuts.

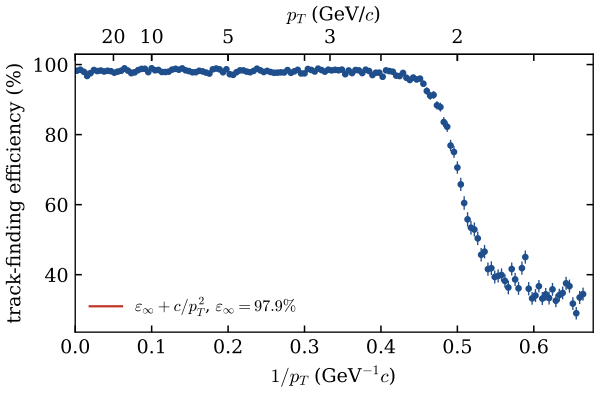


Figure 3: Track-finding efficiency (at least five surviving IT and OT hits) against $1/p_T$, with the fit $\varepsilon_\infty + c/p_T^2$ to the muons above 10 GeV/c.

5 Fake tracks after the selection

The limit. Paper [8] shows that the expected number of combinations of background hits that pass the goodness-of-fit cut of a track fit, with hits on at least k of the layers, is a lower limit

on the number of fake tracks of any reconstruction algorithm that keeps the efficiency for real tracks given by that cut. For layers of any shape and noise densities ρ_i that vary across them,

$$E(\geq k) = \int \mathrm{d}^4\theta \sum_{\substack{S \subseteq C(\theta) \\ |S| \geq k}} V_{n_{\mathrm{dof}}}\, c_S^{n_{\mathrm{dof}}/2} \sqrt{\det\left(J_S^{\mathsf{T}} J_S\right)} \prod_{i \in S} \rho_i\, \sigma^2, \tag{3}$$

where θ are the helix parameters, $C(\theta)$ the layers crossed, J_S the derivatives of the crossing points on the layers of S in units of the resolution, c_S the χ^2 cut and V_d the volume of the unit d-sphere. We require at least five of the six barrel layers, with a one-sided 3σ cut for each set of layers. The integral is evaluated by sampling helices with p_T above 10 GeV/c and the densities after the cuts, taken from the background sample as functions of $|z|$ in each layer; the azimuthal symmetry of the BIB lets us sample a narrow range of azimuth.

The acceptance. The limit refers to a given acceptance: the tracks that the algorithm must find. Real tracks come from the luminous region, and restricting the longitudinal origin of the fitted track removes the fakes that point elsewhere at no cost in efficiency. We take a luminous region of 1.5 mm, the bunch length foreseen at a 10 TeV collider [2], and require $|z_0| < 4.5$ mm for the fitted track. This *track* z_0 should not be confused with the per-hit z_0 of section 3, whose cut is tens of millimetres wide: the fit of five or six hits measures the origin of a real track to 0.1–0.3 mm. Without it the barrel would have 1.1 expected fakes per bunch crossing; with it, 68 times fewer.

Results. With $|z_0| < 4.5$ mm the whole barrel has

$$E(\geq 5) = 1.7 \times 10^{-2} \text{ fakes per bunch crossing}, \tag{4}$$

almost entirely from five-layer combinations: the six-layer term is 2.9×10^{-6}. The fakes concentrate at $|\cot\vartheta|$ between 0.5 and 1 (figure 5). The background density after the cuts is highest at intermediate $|z|$ in every layer, up to 1.4–1.8 times its mean, and these tracks cross the layers there, obliquely. For comparison, one projective tower of six square planes with an outermost plane of 1 m^2, placed at $\vartheta = 90°$, gives 4.1×10^{-3}. Since the five-layer terms dominate, the expected number grows as the fifth power of a common scale factor of the densities; one fake per crossing is reached at 2.3 times the simulated density (figure 4). The number of distinct fakes is smaller than E by the factor $e^{-\eta/2}$, where the window occupancy η measures how often accepted combinations share hits [8]; we expect $\eta \ll 1$ at these densities [compute η for the barrel].

Margin. A factor 2.3 in density is the margin against an underestimate of the BIB in the simulation, against statistical fluctuations of a single simulated bunch crossing, and against everything the limit leaves out (section 7). Equivalently, at the simulated density the limit stays below one per crossing for resolutions up to about twice the assumed ones (section 6).

6 Resolution requirements

Scans. We vary one resolution at a time around the operating point (100 μm, 100 ps, 1.5°), from ten times better to ten times worse, keeping the other two fixed. At every point the time and z_0 cuts are derived again at 98% signal containment, so the efficiency stays constant by construction, and the background densities after the cuts and the fake-track limit are recomputed. Figure 6 shows the limit for one tower and for the whole barrel, scaled from the tower by their ratio at the operating point (4.1), and the track-finding efficiency. The level of one fake per crossing in the whole barrel is reached at a position resolution of 200 μm, a time resolution of 220 ps or a direction resolution of 4.1°.

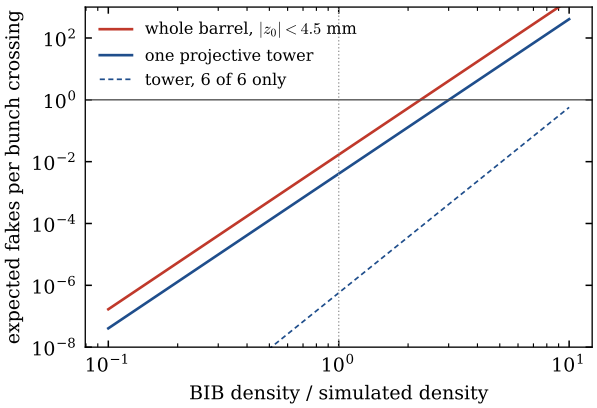


Figure 4: Expected fakes per bunch crossing against a common scale factor of the background densities, for the whole barrel and for one projective tower.

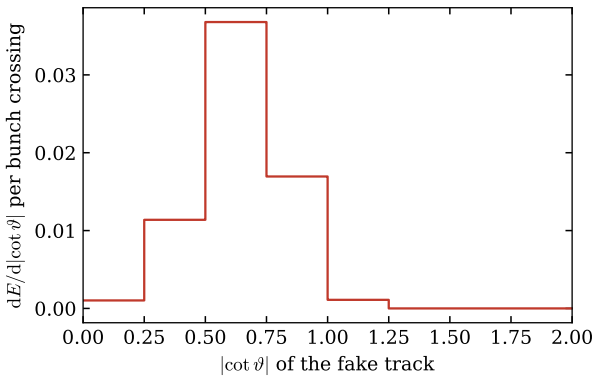


Figure 5: Distribution of the expected fakes of the whole barrel in $|\cot\vartheta|$, with $|z_0| < 4.5$ mm.

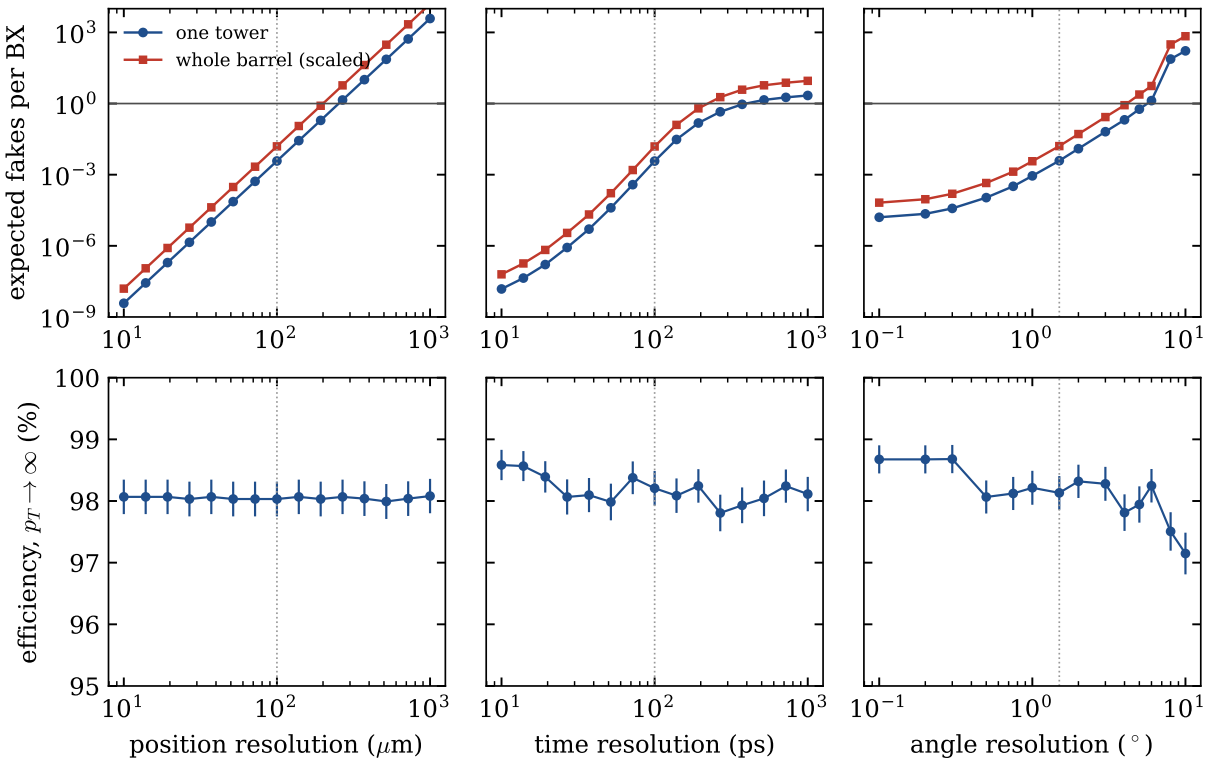


Figure 6: Top: expected fakes per bunch crossing against the position, time and direction resolutions, varied one at a time around the operating point (dotted line), for one projective tower (blue) and for the whole barrel with $|z_0| < 4.5$ mm (red, scaled from the tower by their ratio at the operating point). Bottom: track-finding efficiency for $p_T \to \infty$ of muons confined to the barrel.

Why the curves differ. The position resolution enters twice: through the per-hit observables, whose cuts barely change, and through the χ^2 cut of the track fit, which makes the probability of a five-layer combination scale as σ^6 [8]. The limit therefore falls by six orders of magnitude for a ten-fold improvement, steadily over the whole range. The time and direction resolutions enter only through the cuts, by changing the density of background hits that survive them, raised to the fifth power. The time scan is steep around the operating point, where the time window cuts into the late peak of the background; at a few hundred picoseconds the window contains the whole peak and the curve flattens, the remaining rejection coming from the other two cuts. Below 30 ps the improvement continues as the window closes on the prompt tail of the background. The direction scan is gentler at small angles, where the z_0 window is no longer set by the resolution but by the multiple scattering of the signal muons (10 mm in the IT and 83 mm in the OT at 0.1°), and steepens above 3°; above 6° the z_0 window of the OT becomes so wide that delayed hits of low-momentum muons curling in the field enter the time-cut distribution, the OT time window opens to several nanoseconds and the limit jumps by two orders of magnitude.

Targets for R&D. Table 3 gives the resolution at which the whole barrel reaches given levels of the limit, each resolution varied alone. Present-day values of position and time resolution, 100 μm and 100 ps, leave a margin of about two in background density; reaching 10^{-4} fakes per crossing, a margin of about six, requires about 45 μm or 50 ps. A direction measurement better than about 1.3° is needed for the limit to stay below 10^{-2}; it is the least developed of the three capabilities and the one on which the selection depends most when it is degraded.

Table 3: Resolution at which the expected fakes per bunch crossing in the whole barrel ($|z_0| <$ 4.5 mm, at least five of six layers) reach the given level, with the other two resolutions at their operating values (100 μm, 100 ps, 1.5°).

expected fakes per crossing	1	10^{-2}	10^{-3}	10^{-4}
position (μm)	200	93	63	43
time (ps)	220	94	67	48
direction (deg)	4.1	1.3	0.67	0.21

7 Discussion and limitations

The background sample. The results rest on a single simulated bunch crossing. The densities after the cuts are measured from 15 000 to 41 000 hits per layer. Their Poisson precision is below 1%, but hits produced by the same background particle are correlated, and the limit, through its fifth-power dependence, multiplies any relative error on the densities by five. Independent smearings of the same sample change the tower limit by 8% (from 4.1×10^{-3} to 3.8×10^{-3}). The μ^+ and μ^- samples give consistent densities. [bootstrap over BIB primaries for a proper uncertainty]

What the limit does not include. The limit counts combinations made only of background hits, with Gaussian resolution and no multiple scattering, for tracks above 10 GeV/c from the luminous region. Clusters of several hits from one particle, merged hits, sensor inefficiency and noise, combinations of background with signal hits and correlated background such as low-momentum particles curling in the field can only add fakes or lose real tracks. A real algorithm will produce more fakes than the limit; the limit says how many are unavoidable.

The endcaps. The same method applies to the endcaps and to the transition region, with disks in place of cylinders [8]; the per-hit z_0 is less selective there, because the direction measures the polar angle with a shorter lever arm, and we leave them to future work.

Cut optimization. The cuts are set by signal containment alone. Optimizing them against the background, or combining the three observables into one likelihood, may gain further, at the same efficiency.

8 Conclusions

With a time and a direction measured on every hit, three cuts that test the compatibility of each hit with a particle from the interaction point reduce the beam-induced background in the barrel of the inner and outer trackers of a 10 TeV muon collider by a factor of 57, keeping 97–98% of the signal hits and 98% of the tracks. With present-day position and time resolutions, 100 μm and 100 ps, and a direction resolution of 1.5°, the algorithm-independent lower limit on fake tracks is 1.7×10^{-2} per bunch crossing for the whole barrel, a factor 2.3 in background density below one fake per crossing: the background does not make track reconstruction impossible in principle. The margin grows rapidly with better resolutions, by six orders of magnitude for a ten-fold improvement in position resolution, and the scans give the resolutions needed for any chosen margin. The direction measurement is the most demanding of the three requirements and a natural target for detector R&D.

Acknowledgments

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A Efficiency for $p_T \to \infty$

The hit and track efficiencies are quoted for $p_T \to \infty$, the limit relevant for the high-momentum tracks to which the fake limit refers. For muons above 10 GeV/c the efficiency is fitted with $\varepsilon(p_T) = \varepsilon_\infty + c/p_T^2$, the form expected from multiple scattering and from the p_T-dependent curvature, whose effects on the observables scale as $1/p_T^2$ in the variance. The fit describes the data well (figure 3); for the track efficiency it gives $\varepsilon_\infty = 97.9 \pm 0.2\%$ and $c = 0.60$ GeV^2. [from docs/efficiency_pt_infinity.pdf: justification and checks]

B Code and data availability

The analysis code, the settings files of the runs used and the scripts that make the figures are available in a public repository [9]. [The input samples: on request, or a public location.]

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